Exponential and Logarithmic Derivatives

MATH 1334 October 23, 2026

3.9 Exponential and logarithmic functions

Question. How do we differentiate exponential and logarithmic functions? In particular, how do we find

$$\frac{d}{dx}e^x, \qquad \frac{d}{dx}\ln x, \qquad \frac{d}{dx}x^x?$$

Before answering, we review these functions and define the number e.

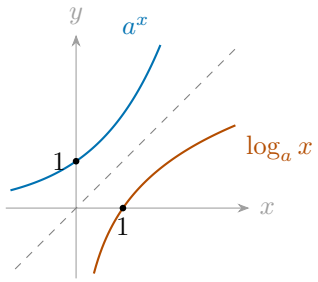
Review

For $a > 0$, let $f(x) = a^x$. The base $a = 1$ gives the constant function $f(x) = 1$. For a nonconstant exponential, the inverse is $f^{-1}(x) = \log_a x$. Thus $y = \log_a x$ means $a^y = x$, and

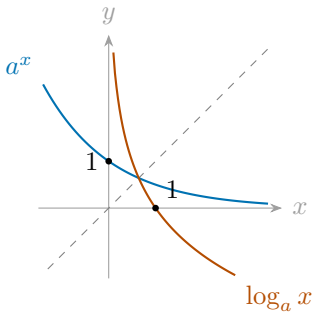
$$a^{\log_a x} = x \quad (x > 0), \qquad \log_a(a^x) = x.$$

A nonconstant exponential has domain $\mathbb{R}$ and range $(0, \infty)$; its logarithmic inverse exchanges the domain and range.

$a > 1$: exponential growth



$0 < a < 1$: exponential decay



The dashed line is $y = x$; each pair of graphs reflects across it.

Algebraic laws

For real r, s and positive M, N,

$$a^{r+s} = a^r a^s, \qquad a^{r-s} = \frac{a^r}{a^s}, \qquad (a^r)^s = a^{rs},$$

$$\log_a(M^r) = r\log_a M, \quad \log_a(MN) = \log_a M + \log_a N, \quad \log_a \frac{M}{N} = \log_a M - \log_a N.$$

Common bases are 2, 10, and $e \approx 2.71828$. For base e, we write $\ln x = \log_e x$, the *natural logarithm*.

What is e?

For $f(x) = a^x$, the slope at $x = 0$, if it exists, is

$$f'(0) = \lim_{h \to 0} \frac{a^{0+h} - a^0}{h} = \lim_{h \to 0} \frac{a^h - 1}{h}.$$

We seek the base for which this slope is 1.

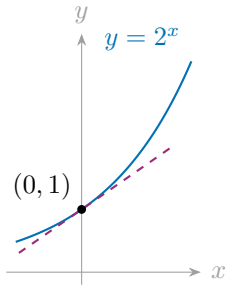
Lemma. *For $a > 0$, the function*

$$F(a) = \lim_{h \to 0} \frac{a^h - 1}{h}$$

is well-defined, continuous, and strictly increasing.

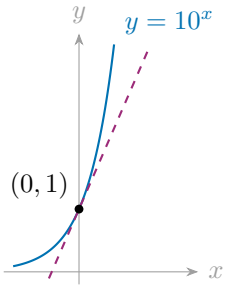
> *Sketch.* Think of $F(a)$ as the tangent slope of $y = a^x$ at $(0, 1)$. The exponential graphs are smooth there, so each has a tangent slope. As the base a changes, the tangent turns continuously; increasing a increases its slope. The two graphs below illustrate this for two bases greater than 1: the larger base gives the steeper tangent.

Smaller base: $a = 2$



Tangent slope: $F(2)$

Larger base: $a = 10$



Tangent slope: $F(10) > F(2)$

Lemma. *There is a unique number $e \in (1, 10)$ such that $F(e) = 1$.*

Proof. Reading the slopes geometrically, $F(1) = 0$, while the graph of 10^x has slope greater than 1 at $(0, 1)$, so $F(10) > 1$. By continuity and the Intermediate Value Theorem, some $e \in (1, 10)$ satisfies $F(e) = 1$. Strict increase makes this e unique. □

Definition. The unique positive real number e for which

$$\lim_{h \to 0} \frac{e^h - 1}{h} = 1$$

is called *Euler's number.*

Note. Another standard, equivalent definition is $e = \lim_{n \to \infty} \left(1 + \frac{1}{n}\right)^n$.

The natural exponential and logarithm

Theorem.

$$(i)\quad \frac{d}{dx}e^x = e^x \quad (x \in \mathbb{R}), \qquad (ii)\quad \frac{d}{dx}\ln x = \frac{1}{x} \quad (x > 0).$$

Proof. **(i)** By the definition of the derivative and the exponent laws,

$$\begin{aligned}\frac{d}{dx}e^x &= \lim_{h\to 0}\frac{e^{x+h}-e^x}{h}\\ &= e^x \lim_{h\to 0}\frac{e^h-1}{h} = e^x \cdot 1 = e^x.\end{aligned}$$

(ii) Let $f(x) = e^x$, so $f^{-1}(x) = \ln x$. By the inverse-function derivative theorem,

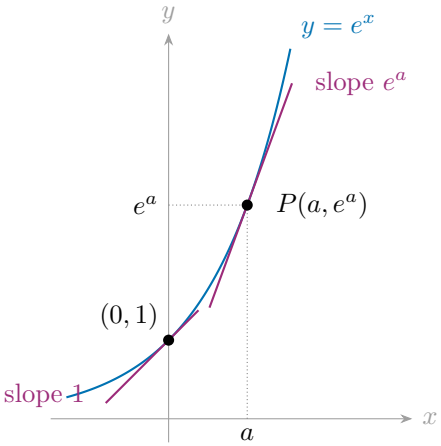
$$\frac{d}{dx}\ln x = (f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))} = \frac{1}{e^{\ln x}} = \frac{1}{x}.$$ □

Geometric meaning

For $f(x) = e^x$, the slope at $P(a, f(a))$ equals the height $f(a)$:

$$f'(a) = f(a) = e^a.$$

At $(0, 1)$ the slope is 1, and the tangent line is $y = 1 + x$.



Relative growth

If $P(t) = P_0 e^{kt}$, with $P_0 > 0$ and k constant, then

$$P'(t) = P_0 e^{kt} k = kP(t), \qquad \frac{P'(t)}{P(t)} = k.$$

The relative growth rate is constant: $k > 0$ gives growth, and $k < 0$ gives decay.

Worked examples

Example 1. Differentiate e^{x^2}.

$$\frac{d}{dx}e^{x^2} = e^{x^2}\,2x = 2xe^{x^2}.$$

Example 2. Differentiate $e^{\cos x}$.

$$\frac{d}{dx}e^{\cos x} = e^{\cos x}\,(-\sin x) = -e^{\cos x}\sin x.$$

Example 3. Differentiate $\ln(2x^3+7)$.

$$\frac{d}{dx}\ln(2x^3+7) = \frac{1}{2x^3+7}\,6x^2 = \frac{6x^2}{2x^3+7}.$$

Practice

Practice 1. Differentiate $e^{\sin x}$.

$$\frac{d}{dx}e^{\sin x} = e^{\sin x}\cos x = e^{\sin x}\cos x.$$

Practice 2. Differentiate $\ln\sqrt{3x-2}$.

Logarithm laws. First simplify the logarithm:

$$\ln\sqrt{3x-2} = \frac{1}{2}\ln(3x-2).$$

Then differentiate:

$$\frac{d}{dx}\ln\sqrt{3x-2} = \frac{1}{2}\cdot\frac{3}{3x-2} = \frac{3}{2(3x-2)}.$$

Chain rule. The same derivative can be found directly:

$$\frac{d}{dx}\ln\sqrt{3x-2} = \frac{1}{\sqrt{3x-2}}\,\frac{1}{2\sqrt{3x-2}}\,3 = \frac{3}{2(3x-2)}.$$

Other bases

Corollary. *For $a > 0$,*

$$(i) \quad \frac{d}{dx} a^x = a^x \ln a.$$

For $a > 0$, $a \neq 1$, and $x > 0$,

$$(ii) \quad \frac{d}{dx} \log_a x = \frac{1}{x \ln a}.$$

Proof. **(i)** Rewrite using e and apply the chain rule:

$$\frac{d}{dx} a^x = \frac{d}{dx} (e^{\ln a})^x = \frac{d}{dx} e^{x \ln a} = e^{x \ln a} \ln a = a^x \ln a.$$

(ii) By change of base, $\log_a x = \ln x / \ln a$. Since $\ln a$ is constant,

$$\frac{d}{dx} \log_a x = \frac{1}{\ln a} \frac{d}{dx} \ln x = \frac{1}{x \ln a}.$$

□

Example 1. Differentiate 2^{3x}.

$$\frac{d}{dx} 2^{3x} = 2^{3x} \ln 2 \cdot 3 = 3(\ln 2) 2^{3x}.$$

Example 2. Differentiate $x^e e^x$. Here e is a constant.

$$\frac{d}{dx} (x^e e^x) = e x^{e-1} e^x + x^e e^x = e^x x^{e-1} (e + x).$$

Example 3. Differentiate $\log_2(6x^4 + 5)$.

$$\frac{d}{dx} \log_2(6x^4 + 5) = \frac{1}{(6x^4 + 5) \ln 2} 24x^3 = \frac{24x^3}{(6x^4 + 5) \ln 2}.$$

Practice

Practice 1. Differentiate $x^\pi \pi^x$.

$$\frac{d}{dx} (x^\pi \pi^x) = \pi x^{\pi - 1} \pi^x + x^\pi \pi^x \ln \pi = x^{\pi - 1} \pi^x (\pi + x \ln \pi).$$

Practice 2. Differentiate $\log_{10}(1 + x^2)$.

$$\frac{d}{dx} \log_{10}(1 + x^2) = \frac{2x}{(1 + x^2) \ln 10}.$$

Logarithmic differentiation

Question. How do we differentiate $f(x)^{g(x)}$, when both the base and the exponent depend on x?

Example. Differentiate x^x.

Method 1: rewrite using e. Since $x^x = e^{x \ln x}$,

$$\begin{aligned}\frac{d}{dx}x^x &= \frac{d}{dx}e^{x \ln x} \\ &= e^{x \ln x}\left(\ln x + x \cdot \frac{1}{x}\right) \\ &= x^x(\ln x + 1).\end{aligned}$$

Method 2: logarithmic differentiation. Let $y = x^x$. Take natural logarithms, then differentiate implicitly:

$$\begin{aligned}\ln y &= x \ln x, \\ \frac{y'}{y} &= \ln x + 1, \\ y' &= y(\ln x + 1) = x^x(\ln x + 1).\end{aligned}$$

The derivative of $\ln y$ is y'/y because y is a function of x.

The general calculation

Suppose f and g are differentiable and $f(x) > 0$. Set $y = f(x)^{g(x)}$. Then $y > 0$, so

$$\begin{aligned}\ln y &= g(x) \ln f(x), \\ \frac{y'}{y} &= g'(x) \ln f(x) + g(x)\frac{f'(x)}{f(x)}, \\ y' &= f(x)^{g(x)}\left(g'(x) \ln f(x) + \frac{g(x)f'(x)}{f(x)}\right).\end{aligned}$$

Taking logarithms turns a variable power into a product. Differentiate that product, solve for y', and substitute the original expression for y.

Examples of logarithmic differentiation

Example 1. Let $y = x^{x^2}$.

$$\begin{aligned} \ln y &= x^2 \ln x, \\ \frac{y'}{y} &= 2x \ln x + x^2 \frac{1}{x} = 2x \ln x + x, \\ y' &= x^{x^2}(2x \ln x + x). \end{aligned}$$

Example 2. Let $y = (x^2 - 4)^{\ln x}$.

$$\begin{aligned} \ln y &= (\ln x) \ln(x^2 - 4), \\ \frac{y'}{y} &= \frac{\ln(x^2 - 4)}{x} + \ln x \cdot \frac{2x}{x^2 - 4}, \\ y' &= (x^2 - 4)^{\ln x} \left(\frac{\ln(x^2 - 4)}{x} + \frac{2x \ln x}{x^2 - 4} \right). \end{aligned}$$

Example 3. Let $y = x^{-1/2}(x^2 + 1)^{2/3}(2x - 3)^4$. Taking logarithms gives

$$\ln y = -\frac{1}{2} \ln x + \frac{2}{3} \ln(x^2 + 1) + 4 \ln |2x - 3|.$$

Differentiate term by term:

$$\frac{y'}{y} = -\frac{1}{2x} + \frac{4x}{3(x^2 + 1)} + \frac{8}{2x - 3}.$$

Thus,

$$y' = x^{-1/2}(x^2 + 1)^{2/3}(2x - 3)^4 \left(-\frac{1}{2x} + \frac{4x}{3(x^2 + 1)} + \frac{8}{2x - 3} \right).$$

Practice

Practice 1. Differentiate $y = x^{\sqrt{2x}}$.

$$\begin{aligned} \ln y &= \sqrt{2x}\ln x, \\ \frac{y'}{y} &= \frac{\ln x}{\sqrt{2x}} + \frac{\sqrt{2x}}{x} = \frac{\ln x + 2}{\sqrt{2x}}, \\ y' &= \frac{x^{\sqrt{2x}}(\ln x + 2)}{\sqrt{2x}}. \end{aligned}$$

Practice 2. Differentiate $y = 5^{\sin(2x)}$.

$$\begin{aligned} y' &= 5^{\sin(2x)}\ln 5 \cdot \cos(2x) \cdot 2 \\ &= 2(\ln 5)\, 5^{\sin(2x)}\cos(2x). \end{aligned}$$

Practice 3. Differentiate $y = (x^2-1)^{\sqrt{x}}$.

$$\begin{aligned} \ln y &= \sqrt{x}\ln(x^2-1), \\ \frac{y'}{y} &= \frac{\ln(x^2-1)}{2\sqrt{x}} + \sqrt{x}\,\frac{2x}{x^2-1}, \\ y' &= (x^2-1)^{\sqrt{x}}\left(\frac{\ln(x^2-1)}{2\sqrt{x}} + \frac{2x\sqrt{x}}{x^2-1}\right). \end{aligned}$$

The power rule for real exponents

Theorem. *For any fixed real number n and $x > 0$,*

$$\frac{d}{dx}x^n = nx^{n-1}.$$

Proof. Let $y = x^n$. Taking logarithms and differentiating gives

$$\ln y = n\ln x, \qquad \frac{y'}{y} = \frac{n}{x}.$$

Therefore $y' = y(n/x) = nx^{n-1}$. □

The restriction $x > 0$ makes this proof work for every real exponent. The rule also holds at other inputs when the particular power function is differentiable there.